

Bug Fix Impact Analysis

METIMMOX-1 Economic Evaluation

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1 Summary

This report compares cost-effectiveness analysis results before and after addressing GitHub issue #118. The comparison assesses whether bug fixes or model changes affected results.

Current Comparison:

- **Issue:** #118 - No title available
- **Baseline:** Commit 4e55627 (2026-02-17 16:57:09)
- **Fixed:** Commit 10df9fe (2026-02-17 16:57:47)
- **Models included:** Single model (legacy snapshot)

2 Metadata Comparison

Table 1: Metadata Comparison

Item	Baseline	Fixed
Git Commit	4e55627	10df9fe
Timestamp	2026-02-17 16:57:09	2026-02-17 16:57:47
R Version	R version 4.3.2 (2023-10-31 ucrt)	R version 4.5.1 (2025-06-13 ucrt)
Issue Number	118	118
Model Structure	0 (joint)	0 (joint)
Multi-Model Snapshot	No	No

2.1 Global Parameters

Table 2: Global Parameter Comparison

	Parameter	Baseline	Fixed	Changed
WTP	WTP	51,000	51,000	No
time_horizon	time_horizon	520	520	No
n_samples	n_samples	5,000	5,000	No
n_sim	n_sim	5,000	5,000	No
cl	cl	0.01923077	0.01923077	No
dr	dr	0.04	0.04	No
DSA_mult	DSA_mult	0.2	0.2	No
USE_BOTH_MODELS	USE_BOTH_MODELS	0	0	No

3 Per-Model Comparisons

3.1 Single Model

3.1.1 Base Case Results

Table 3: Single Model - Incremental Cost Comparison

Strategy	Baseline (EUR)	Fixed (EUR)	Difference (EUR)
control	0.00	0.00	0.00
crp	34,003.49	34,003.49	0.00
tlr	64,079.67	64,079.67	0.00
tmb_braf	42,665.89	42,665.89	0.00

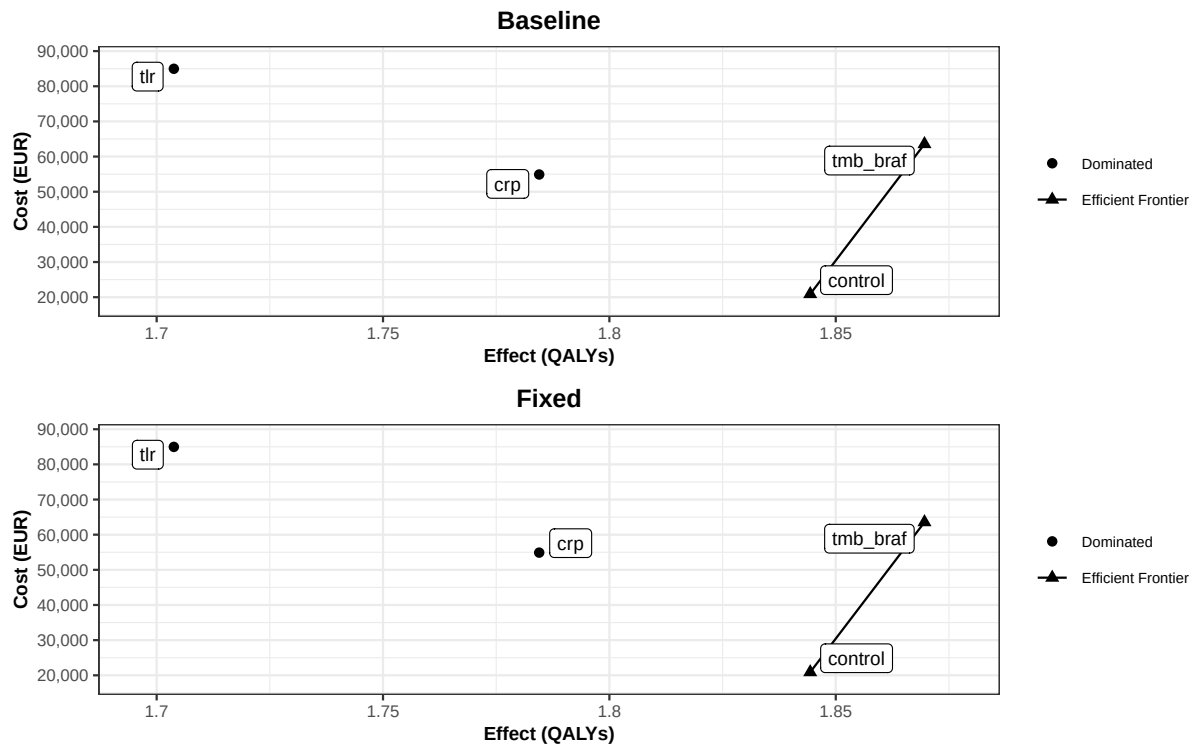
Table 4: Single Model - Incremental Effect Comparison

Strategy	Baseline (QALYs)	Fixed (QALYs)	Difference (QALYs)
control	0.0000	0.0000	0.0000
crp	-0.0599	-0.0599	0.0000
tlr	-0.1406	-0.1406	0.0000
tmb_braf	0.0252	0.0252	0.0000

Table 5: Single Model - ICER Comparison

Strategy	Baseline (EUR/QALY)	Fixed (EUR/QALY)	Difference (EUR/QALY)
control	—	—	—
crp	—	—	—
tlr	—	—	—
tmb_braf	1,690,480	1,690,480	0

3.1.2 Efficiency Frontier



TableGrob (2 x 1) “arrange”: 2 grobs z cells name grob 1 1 (1-1,1-1) arrange gtable[layout] 2 2 (2-2,1-1) arrange gtable[layout]

3.1.3 Net Monetary Benefit

Table 6: Single Model - NMB at WTP = EUR 51,000

	Strategy	Baseline (EUR)	Fixed (EUR)	Difference (EUR)
1	control	73,168.05	73,168.05	0.00
2	crp	36,111.38	36,111.38	0.00
4	tmb_braf	31,789.34	31,789.34	0.00
3	tlr	1,919.73	1,919.73	0.00

Optimal Strategy: Unchanged (control)

3.1.4 PSA Results

Table 7: Single Model - PSA Cost Results

Strategy	Baseline Mean (2.5%; 97.5%)	Fixed Mean (2.5%; 97.5%)	Difference (EUR)
control	EUR 21,159.71 (EUR 16,050.79; EUR 27,214.85)	EUR 21,074.25 (EUR 15,975.03; EUR 27,111.55)	-85.46
crp	EUR 54,853.90 (EUR 39,402.67; EUR 74,501.34)	EUR 55,058.21 (EUR 39,626.84; EUR 75,964.95)	204.31
tlr	EUR 85,034.33 (EUR 56,367.75; EUR 121,802.72)	EUR 85,333.78 (EUR 56,282.73; EUR 122,389.18)	299.45
tmb_braf	EUR 63,354.14 (EUR 44,780.85; EUR 88,636.87)	EUR 63,490.77 (EUR 45,377.07; EUR 88,712.14)	136.63

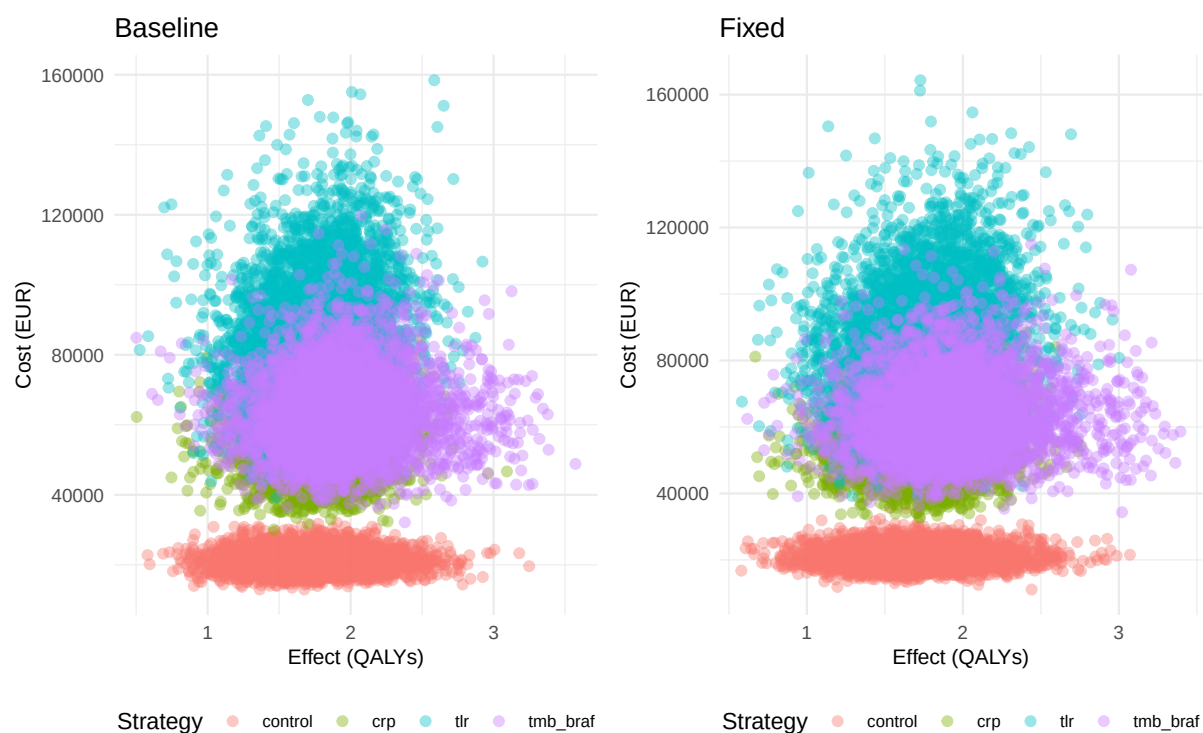
Table 8: Single Model - PSA Effect Results

Strategy	Baseline Mean (2.5%; 97.5%)	Fixed Mean (2.5%; 97.5%)	Difference (QALYs)
control	1.7077 (1.1049; 2.3999)	1.6988 (1.0762; 2.3515)	-0.0090
crp	1.8148 (1.2589; 2.3106)	1.8097 (1.2080; 2.3102)	-0.0051
tlr	1.7582 (1.1441; 2.3575)	1.7537 (1.1176; 2.3700)	-0.0044
tmb_braf	1.9382 (1.2604; 2.7930)	1.9296 (1.2485; 2.7611)	-0.0086

Table 9: Single Model - PSA ICER Results

Strategy	Baseline Mean ICER	Fixed Mean ICER
control	—	—
crp	EUR 314,680/QALY	EUR 306,223/QALY
tlr	EUR 1,266,366/QALY	EUR 1,168,515/QALY
tmb_braf	EUR 183,130/QALY	EUR 183,760/QALY

3.1.5 PSA Scatter Plots



TableGrob (1 x 2) “arrange”: 2 grobs z cells name grob 1 1 (1-1,1-1) arrange gtable[layout] 2 2 (1-1,2-2) arrange gtable[layout]

4 Cross-Model Summary

Single-model snapshot. See the detailed comparison above.

5 Comparison Across Recent Issues

5.1 Available Snapshot Pairs

Table 10: All Available Snapshot Pairs

Issue		Title	Baseline Commit	Fixed Commit
1	118	—	4e55627	10df9fe
2	18	—	71f8230	71f8230
18	99	—	376b4c6	d1eef51
17	97	—	457533e	457533e
5	47	—	2411bdb	2411bdb
4	46	—	e8b3c27	2d5181c
16	94	—	e77415f	3f98095
11	75	—	2b04ea7	c15aa47
15	90	—	9e1b590	7d68bca
12	76	—	64cf543	64cf543
14	87	—	42d4acb	0a15902
13	86	—	ee96c31	f3e5604
10	70	—	998dacd	605ff90
9	68	—	4665f93	998dacd
3	42	—	607b539	cadf81a
8	67	—	f3674e8	d5d1356
6	62	—	d046c7c	f3674e8
7	64	—	8681d1e	8681d1e

5.2 Multi-Issue Impact Summary

This table summarizes the impact of bug fixes across the most recent issues:

Table 11: Summary of Bug Fix Impacts (5 Most Recent Issues)

Issue	Title	Opt. (BL)	Opt. (Fix)	Chg?	Max Cost Delta (EUR)	Max Effect Delta (QALYs)	Max ICER Delta (EUR/QALY)
118	—	control	control	No	0.00	0.0000	0
18	—	control	control	No	1,314.18	0.0000	44,593
99	—	control	control	No	0.00	0.0000	0
97	—	control	control	No	0.00	0.0000	0
47	—	control	control	No	0.00	0.0000	0

6 Summary

The comparison for issue #118 shows:

- **Models compared:** Single model (legacy snapshot)
- **Optimal Strategy (Single Model):** Unchanged (control)
- **Maximum Cost Change:** EUR 0.00 (minor)
- **Maximum Effect Change:** 0.000 QALYs (minor)
- **Maximum ICER Change:** EUR 0/QALY (minor)
- **Parameter Changes:** None detected

Interpretation: Changes are within expected bounds for bug fixes that improve numerical stability or code quality.

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Report version: 3.0